

Motor Bearing Condition Monitoring from Vibration Signals

Physics-grounded condition monitoring with the Case Western Reserve University bearing dataset

8	40	1 Hz	82.5%
source records	one-second windows	FFT bin spacing	exploratory cross-load ML

Problem and physical mechanism

A localized race or rolling-element defect produces a short impact when it enters contact in the loaded zone. The impact excites a higher-frequency structural resonance, creating a decaying burst. Repeated bursts form an amplitude-modulated signal: the resonance is the carrier and the defect encounter rate is the modulation frequency. Time statistics can flag impulsiveness; a Hilbert-envelope spectrum can recover the component-specific repetition rate.

Dataset scope Eight official drive-end records: normal, 0.007-inch inner-race, ball, and outer-race faults at 0 and 3 HP. The outer-race fault is centered at 6 o'clock in the load zone. Five balanced one-second windows are retained per record.	Processing chain SHA-256 verification -> DC removal -> anti-alias rate standardization -> time features -> Hann-windowed amplitude-correct RFFT -> bearing kinematics -> fixed 2-5 kHz bandpass -> Hilbert envelope -> cross-load evaluation.
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Bearing impacts revealed by envelope demodulation

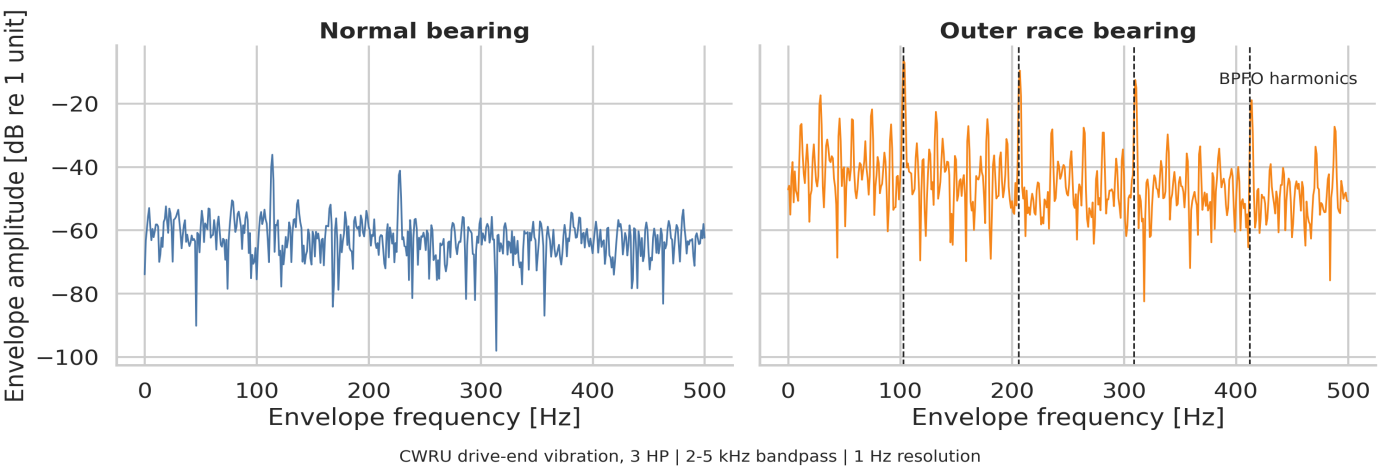


Figure 1. Envelope demodulation suppresses carrier complexity and exposes BPFO harmonics in the outer-race case. Amplitude is reported in CWRU units because public dataset pages do not establish absolute calibration.

Method validity, results, and deployment judgment

Key equations and sampling constraints

Indicator	Definition / engineering meaning
Frequency resolution	$\Delta f = f_s / N = 12,000 / 12,000 = 1 \text{ Hz}$. Nyquist = 6 kHz.
Race frequencies	$BPFO = (n/2) f_r [1 - (d/D) \cos(\theta)]$; BPFI uses the plus sign.
Ball and cage	$BSF = (D/2d) f_r [1 - ((d/D) \cos(\theta))^2]$; $FTF = 0.5 f_r [1 - (d/D) \cos(\theta)]$.
Envelope	$z[n] = x_{BP}[n] + j H\{x_{BP}[n]\}$; $e[n] = z[n] $. $FFT\{e[n] - \text{mean}(e)\}$ reveals impact repetition.

Cross-load experiment

Fault / expected family	0 HP median	3 HP median	Interpretation
Outer race / BPFO	40.0 dB	36.1 dB	Strong at both loads
Inner race / BPFI	34.8 dB	35.0 dB	Stable across load
Ball / BSF	5.8 dB	9.9 dB	Weaker, load-sensitive

The load change shifts shaft speed, so theoretical frequencies are recalculated from each file's measured RPM. Race-fault evidence remains strong; BSF is near the 6 dB reference at 0 HP and improves at 3 HP. The bidirectional load-held-out logistic regression reaches 82.5%, but this is not a production accuracy claim: one recording represents each class/load.

Reliability and deployment considerations

- EDM-seeded, isolated laboratory faults do not represent natural progression or multiple simultaneous defects.
- Normal and fault records have different source sampling histories; resampling creates a common grid, not identical acquisition chains.
- Ideal no-slip kinematics require tolerance for slip and shaft-speed sidebands in field data.
- Sensor mounting, orientation, dynamic range, anti-aliasing, cabling, and structural transfer paths affect repeatability.
- Production alarms require asset-specific baselines, speed/load context, persistence logic, sensor-health checks, and maintenance confirmation.

Verification and sources

Automated tests verify selected signal lengths, the 12 kHz analysis rate, FFT peak location and amplitude, Nyquist endpoint, segmentation, and feature calculations. Sources: CWRU Bearing Data Center, Apparatus & Procedures, Bearing Information, and official data tables at engineering.case.edu/bearingdatacenter. Analysis and code are independently authored; advanced methods are intentionally outside the MVP.